

ROBUST STATISTICAL CHANGE DETECTION OF COMPLEX-VALUED VAE RECONSTRUCTION FOR SAR ANOMALY DETECTION

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ABSTRACT

The lack of labeled data for anomaly detection in SAR imaging motivates the use of unsupervised approaches. Several data-driven methodologies based on reconstruction errors have been proposed to address this issue. Moreover, the heterogeneous nature of SAR clutter makes anomaly detection challenging and can increase the false alarm rate. In this paper, we propose a SAR anomaly detection framework based on a complex-valued Variational Autoencoder, which recasts anomaly detection as a change-detection problem using complex-valued, robust-statistics-based change detectors. The proposed detectors account for the heterogeneous and non-Gaussian nature of SAR clutter, providing improved robustness to false alarms. Experimental results show that the proposed approach outperforms reconstruction-based methods with conventional detectors, particularly in heterogeneous clutter.

Index Terms— Anomaly detection, SAR, Complex-valued Neural Networks, GLRT, Complex Compound Gaussian

1. INTRODUCTION

Anomaly detection aims to detect targets with an unknown signature (steering vector) that deviates from the statistical characteristics of a background. In SAR imagery, existing approaches range from statistical detectors such as the Reed-Xiaoli Detector (RXD) [1] to machine learning-based reconstruction methods using Autoencoder architectures [2, 3, 4, 5, 6, 7]. In the latter case, anomalies are typically identified from their reconstruction errors. Nevertheless, these errors are affected not only by anomalous content but also by reconstruction quality and the high variability of SAR imagery. Consequently, directly thresholding the reconstruction error may result in numerous false alarms, particularly in heterogeneous backgrounds.

The increasing spatial resolution of SAR sensors further underscores these challenges by providing finer-scale observations of complex, heterogeneous scenes. Historically, many anomaly detection approaches assume Gaussian clutter [8]; however, SAR data often exhibit heterogeneous and heavy-tailed statistics that can violate this assumption. Several studies [9, 10, 11] have highlighted that the complex Gaussian distribution may no longer be adequate for characterizing high-resolution SAR images, whose statistics are strongly influenced by texture and exhibit heavy tails.

To better account for these properties, numerous statistical models have been proposed, notably the K and Weibull distributions. Despite their differences in formulation, these models fall within a common probabilistic framework: that of *Complex Elliptic Symmetry* (CES) distributions. This family of distributions constitutes a unifying model particularly well-suited for representing modern SAR data, modeling both Gaussian behavior and heavy-tailed distributions, as notably demonstrated in the work of [12].

CES models have been extensively studied in the literature and successfully applied to various SAR applications, including *target detection* [13], *change detection* [14, 15, 16], and SAR data modeling [12]. Motivated by these applications, we adopt a change-detection perspective for anomaly detection, treating the input image and its VAE reconstruction as two observations of the same region. Anomalies are then characterized as changes between these two observations.

This paper studies the contribution of robust statistics in detecting anomalies in high-resolution SAR images with a Complex-valued VAE and change detection. Essentially, we push further the initial anomaly detection proposition in [5] with *Generalized Likelihood Ratio Test* (GLRT)-based detectors derived from the *Complex Compound Gaussian* (CCG) distribution proposed in [16], in a heterogeneous scenario. Finally, the detection performance will be discussed.

Notations: Matrices are in bold and capital, vectors in bold. For any matrix $\mathbf{A}$ or vector, $\mathbf{A}^H$ is the Hermitian transpose of $\mathbf{A}$. The symbols $\odot$ and $\oslash$ denote the Hadamard element-wise product and division, respectively. On vectors, the operator $\circ$ acts on each of its components (power, log, modulus).

2. RELATED WORK

Reed-Xiaoli Detector. In the polarimetric setting (HH, HV, VV), $\mathbf{x}_t \in \mathbb{C}^3$ denotes the test pixel, while $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_N) \in \mathbb{C}^{3 \times N}$ contains the centered secondary data used to estimate the clutter covariance matrix. The RXD [1], formulated within the GLRT framework, measures the deviation of the test pixel from the estimated background using the Mahalanobis distance:

$$RXD(\mathbf{x}_t) = \mathbf{x}_t^H \hat{\Sigma}_{SCM}^{-1} \mathbf{x}_t \stackrel{H_1}{\underset{H_0}{\gtrless}} \lambda, \quad (1)$$

where $\hat{\Sigma}_{SCM} = \mathbf{X} \mathbf{X}^H / N$ is the Sample Covariance Matrix estimated from the secondary data. However, the RXD is derived under the Gaussian assumption, which is inadequate for the heterogeneous and heavy-tailed statistics of SAR data. Robust estimators such as the Tyler estimator [17] address this limitation by estimating the covariance structure without requiring an explicit parametric clutter distribution. This estimator was investigated for anomaly detection in [18].

Reconstruction based methodologies. Rather than specifying a background distribution, reconstruction-based methods learn representations of normal data and use reconstruction errors to detect anomalies. [19, 3, 2, 5] used Convolutional Autoencoders to model SAR clutter and detect poorly reconstructed observations. Muzeau et al. [2, 3] further incorporated the despeckling process to reduce speckle-induced false alarms. The latest effort [5] exploits a complex-valued Variational Autoencoder to learn the anomaly-free distribution of SAR images

Change Detection. Muzeau et al. proposed a Frobenius-based change detector [3]. From a statistical perspective, however, the Frobenius norm can be interpreted as a particular case of a maximum-likelihood change detector under the assumption of white, isotropic Gaussian noise. More general statistical formulations have therefore been investigated, including the Novak detector [20], which explicitly models the data according to Gaussian statistics under the GLRT framework between two SAR images taken on dates $t = 1$ and $t = 2$:

$$\hat{\Lambda}_{\text{Novak}} = \frac{\left| \frac{1}{2} \sum_{t=1}^2 \hat{\Sigma}_{SCM}^t \right|^{2N}}{\prod_{t=1}^2 \left| \hat{\Sigma}_{SCM}^t \right|^N} \underset{H_0}{\overset{H_1}{\geq}} \lambda, \quad (2)$$

Nevertheless, the non-Gaussianity and heavy-tailed nature of SAR clutter may deteriorate the performance of such parametric approaches. To alleviate this limitation, Mian et al. [21, 16] proposed robust change detectors within the CCG distribution, thereby providing improved resilience to heterogeneous and non-Gaussian clutter distributions.

3. ANOMALY DETECTION SCHEME

In SAR imaging, we consider the challenging setting in which anomalies are unlabeled, and the training data may be contaminated by anomalous observations. A complex-valued Variational AutoEncoder (VAE) is trained on data containing both regular observations and potential anomalies. We expect the network to reconstruct regular observations more accurately than anomalous ones, resulting in larger reconstruction discrepancies for anomalies. The latter are then interpreted as changes between the input and its reconstruction and used to generate anomaly maps through advanced statistical change detectors.

3.1. Complex-valued Variational Autoencoder

We adopt a complex-valued β -VAE [22] specifically tailored to Single Look Complex (SLC) data. This extension of the classical VAEs proposed in [23, 24] to the complex domain allows direct exploitation of the inherently complex nature of SAR signals while preserving relevant statistical properties of the radar data, such as non-circularity and intrinsic magnitude-phase relationships. The overall architecture is depicted in Fig. 1.

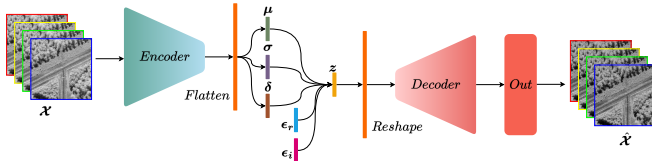


Fig. 1: Complex-Valued Variational Autoencoder.

Given an input image $\mathcal{X} \in \mathbb{C}^{3 \times h \times w}$, the 3 polarization channels of the complex-valued SAR, the encoder spatially compresses into a latent representation $\mathbf{z} \in \mathbb{C}^q$. Each encoding block consists of two stacked sequences *Convolution–BatchNorm–Activation*. It then outputs the mean $\boldsymbol{\mu} \in \mathbb{C}^q$, the variance $\boldsymbol{\sigma}^{\circ 2} \in \mathbb{R}^{q+}$ and the pseudo-covariance $\boldsymbol{\delta} \in \mathbb{C}^q$ to sample the latent space $\mathbf{z}$ and ensuring the gradient flow via the reparameterization trick. For the complex-valued case, the latter has been adapted by Nakashika [23] and further developed in [24]:

$$\mathbf{z} = \boldsymbol{\mu} + \mathbf{k}_r \odot \boldsymbol{\epsilon}_r + i \mathbf{k}_i \odot \boldsymbol{\epsilon}_i, \quad (3)$$

where $\begin{cases} \mathbf{k}_r = 2^{-\frac{1}{2}} (\boldsymbol{\mu} + \boldsymbol{\delta}) \odot (\boldsymbol{\sigma} + \text{Re}(\boldsymbol{\delta}))^{\circ \frac{1}{2}}, \\ \mathbf{k}_i = 2^{-\frac{1}{2}} (\boldsymbol{\sigma}^{\circ 2} - (|\boldsymbol{\delta}|^{\circ 2})^{\circ 2})^{\circ \frac{1}{2}} \odot (\boldsymbol{\sigma} + \text{Re}(\boldsymbol{\delta}))^{\circ \frac{1}{2}} \end{cases}$

The decoder reconstructs the input image by symmetrically mirroring the encoder's architecture. Oversampling is performed using complex transposed convolution layers. We denote $\hat{\mathcal{X}} \in \mathbb{C}^{3 \times h \times w}$ the reconstructed image.

Optimizing a β -VAE involves minimizing the *Evidence Lower Bound* loss function:

$$\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \beta D_{KL}. \quad (4)$$

The variational objective comprises a reconstruction loss, which ensures high-quality image reconstruction, and the Kullback-Leibler (KL) divergence, which enforces a proximity between the latent space distribution and the desired prior, expressed in [24]:

$$D_{KL} = \|\boldsymbol{\mu}\|^2 + \mathbf{1}_q^T \left(\boldsymbol{\sigma} - \mathbf{1} - \frac{1}{2} \log^{\circ} (\boldsymbol{\sigma}^{\circ 2} - (|\boldsymbol{\delta}|^{\circ 2})^{\circ 2}) \right). \quad (5)$$

3.2. Change Detection

Robust change detection in SAR time series has been investigated by Mian et al. [16] within the GLRT framework, using the CCG distribution, a method known as *covariance equality testing*. To cast anomaly detection within the change detection framework, one can treat an input and its reconstruction from a VAE as two observations of the same region and define anomalies as changes between them. The resulting change detection process is illustrated in Fig. 2.

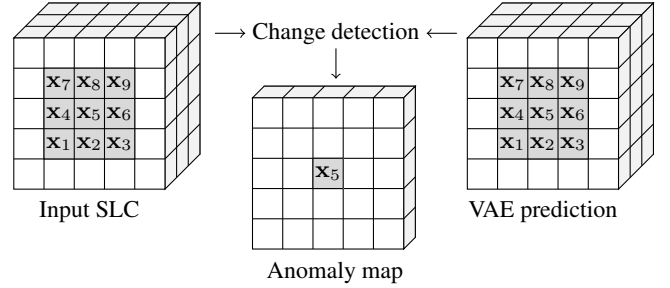


Fig. 2: Change detection between the SLC input and the VAE prediction. The shaded area defines the *boxcar* data used to estimate the covariance matrices to build a test pixel in the anomaly map.

The stochastic representation of the *CES* distribution is closely related to the *Complex Compound Gaussian* - CCG model, where the covariance matrix is parameterized as $\boldsymbol{\Sigma} = \tau \mathbf{V}$. Here, τ denotes the *scale parameter*, which controls the pixel amplitudes and may vary with the spatial location, while $\mathbf{V}$ denotes the *shape matrix*, which characterizes the local structure and heterogeneity of the clutter. Both τ and $\mathbf{V}$ depend on the boxcar data over which the parameters are estimated.

For each observation, whether the input image ($t = 1$) or the VAE's prediction ($t = 2$), let $\mathbf{x}_k^t$ be the test pixel at position k . The probability density function of the CCG model for each $\mathbf{x}_k^t \sim \mathcal{CN}(\mathbf{0}, \tau_k^t \mathbf{V}_t)$ is:

$$p(\mathbf{x}_k^t) = \frac{1}{\pi^p |\tau_k^t \mathbf{V}_t|} \exp \left(-\frac{(\mathbf{x}_k^t)^H \mathbf{V}_t^{-1} \mathbf{x}_k^t}{\tau_k^t} \right), \quad (6)$$

with $\mathbf{V}$ estimated via the pixels of the boxcar area.

Mathematically, the covariance equality testing can be written as a binary hypothesis problem of the following form:

$$\begin{cases} H_0 : & \boldsymbol{\theta}_1 = \boldsymbol{\theta}_2 = \boldsymbol{\theta}_0 \quad \& \quad \boldsymbol{\Phi}_1, \boldsymbol{\Phi}_2, \\ H_1 : & \boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2 \quad \& \quad \boldsymbol{\Phi}_1, \boldsymbol{\Phi}_2 \end{cases}, \quad (7)$$

with $\boldsymbol{\theta}$ is either covariance, scale parameter only, or shape matrix. The remaining unknown parameters, or nuisance variables, are denoted by $\boldsymbol{\Phi}$, and can be empty or consist of scale and/or shape parameters, depending on the definition of $\boldsymbol{\theta}$.

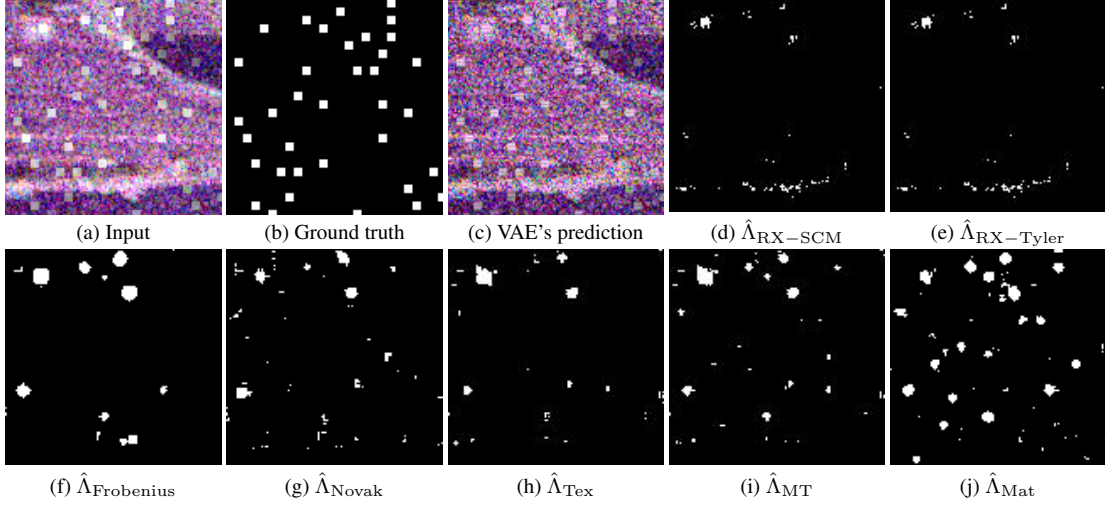


Fig. 3: Experiments on synthetic anomalies. Detection maps generated are displayed at $P_{fa} = 0.5\%$.

Under this binary hypothesis test, the associated GLRT is formulated below, with $\mathcal{X}_t = [\mathbf{x}_1^t, \dots, \mathbf{x}_N^t] \in \mathbb{C}^{3 \times N}$, $t \in \{1, 2\}$:

$$\hat{\Lambda} = \frac{\max_{\theta_1, \theta_2, \Phi_1, \Phi_2} p(\mathcal{X}_1, \mathcal{X}_2; \theta_1, \theta_2, \Phi_1, \Phi_2 | H_1)}{\max_{\theta_0, \Phi_1, \Phi_2} p(\mathcal{X}_1, \mathcal{X}_2; \theta_0, \Phi_1, \Phi_2 | H_0)} \quad (8)$$

When testing the shape matrix only, meaning $\theta_t = \{\mathbf{V}_t\}$ and $\Phi_t = \{\tau_1^t, \dots, \tau_N^t\}$, deriving Eq. 8 with Eq. 6 gives the Mat detector [16], expressed as:

$$\hat{\Lambda}_{Mat} = \frac{|\hat{\mathbf{V}}_{Mat}^0|^{2N} \prod_{t=1}^2 \prod_{k=1}^N \frac{(\mathbf{x}_k^t)^H (\hat{\mathbf{V}}_{Mat}^0)^{-1} \mathbf{x}_k^t}{(\mathbf{x}_k^t)^H (\hat{\mathbf{V}}_{TE}^t)^{-1} \mathbf{x}_k^t}}{\prod_{t=1}^2 |\hat{\mathbf{V}}_{TE}^t|^N} \stackrel{H_1}{\underset{H_0}{\gtrless}} \lambda, \quad (9)$$

with $\hat{\mathbf{V}}_{Mat}^0 = \frac{p}{2N} \sum_{k=1}^N \frac{\mathbf{x}_k^t (\mathbf{x}_k^t)^H}{(\mathbf{x}_k^t)^H (\hat{\mathbf{V}}_{Mat}^0)^{-1} \mathbf{x}_k^t}$ and the Tyler M-estimated covariance matrix at time t :

$$\hat{\mathbf{V}}_{TE}^t = \frac{p}{N} \sum_{k=1}^N \frac{\mathbf{x}_k^t (\mathbf{x}_k^t)^H}{(\mathbf{x}_k^t)^H (\hat{\mathbf{V}}_{TE}^t)^{-1} \mathbf{x}_k^t}. \quad (10)$$

To detect changes when the scale parameter varies only, $\theta_t = \{\tau_1^t, \dots, \tau_N^t\}$ and $\Phi_t = \{\mathbf{V}_t\}$, the GLRT ratio becomes the Robust Tex [16]:

$$\hat{\Lambda}_{Tex} = \prod_{t=1}^2 \frac{|\hat{\mathbf{V}}_{Tex}^t|^N}{|\hat{\mathbf{V}}_{TE}^t|^N} \prod_{k=1}^N \frac{\left(\sum_{t=1}^2 (\mathbf{x}_k^t)^H (\hat{\mathbf{V}}_{Tex}^t)^{-1} \mathbf{x}_k^t \right)^{2p}}{2^{2p} \prod_{t=1}^2 \left((\mathbf{x}_k^t)^H (\hat{\mathbf{V}}_{TE}^t)^{-1} \mathbf{x}_k^t \right)^p} \stackrel{H_1}{\underset{H_0}{\gtrless}} \lambda, \quad (11)$$

$$\text{with } \hat{\mathbf{V}}_{Tex}^t = \frac{2p}{N} \sum_{k=1}^N \frac{\mathbf{x}_k^t (\mathbf{x}_k^t)^H}{\sum_{t=1}^2 (\mathbf{x}_k^t)^H (\hat{\mathbf{V}}_{Tex}^t)^{-1} \mathbf{x}_k^t}.$$

Finally, Eq. 8 ratio under the shape matrix and scale parameter test scheme takes the form of the Robust MT detector [16]:

$$\hat{\Lambda}_{MT} = \frac{|\hat{\mathbf{V}}_0^{MT}|^{2N} \prod_{k=1}^N \left(\sum_{t=1}^2 (\mathbf{x}_k^t)^H (\hat{\mathbf{V}}_0^{MT})^{-1} \mathbf{x}_k^t \right)^{2p}}{\prod_{t=1}^2 |\hat{\mathbf{V}}_{TE}^t|^N \prod_{k=1}^N \left(\sum_{t=1}^2 (\mathbf{x}_k^t)^H (\hat{\mathbf{V}}_{TE}^t)^{-1} \mathbf{x}_k^t \right)^p} \stackrel{H_1}{\underset{H_0}{\gtrless}} \lambda, \quad (12)$$

$$\text{with } \hat{\mathbf{V}}_0^{MT} = \frac{p}{N} \sum_{k=1}^N \frac{\sum_{t=1}^2 (\mathbf{x}_k^t)^H \mathbf{x}_k^t}{\sum_{t=1}^2 (\mathbf{x}_k^t)^H (\hat{\mathbf{V}}_0^{MT})^{-1} \mathbf{x}_k^t}.$$

4. RESULTS

4.1. Synthetic data

We perform quantitative evaluation using synthetic anomalies (see Fig. 3a) with amplitude A injected into a heterogeneous clutter regions at 15 dB of signal-to-clutter ratio (SCR) given by $A^2 \mathbf{p}^H \hat{\mathbf{\Sigma}}_{SCM}^{-1} \mathbf{p}$ where $\mathbf{p} = (1, 1, 1)^T$ represents the steering vector and $\hat{\mathbf{\Sigma}}_{SCM}$ is the estimated covariance matrix of the local clutter surrounding the target. Note that the anomaly sparsity hypothesis is only needed for the training of the VAE.

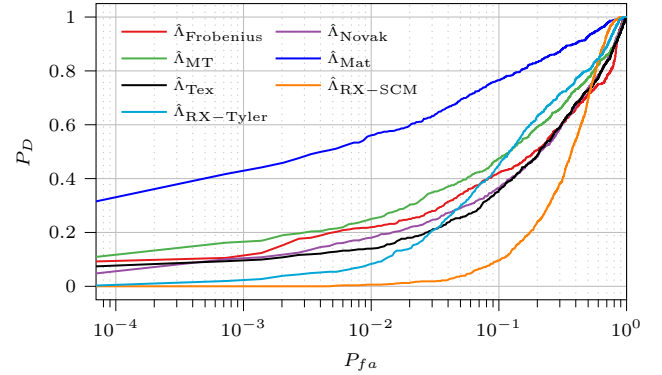


Fig. 4: Probability of Detection versus $\log_{10}(P_{fa})$ for an anomaly with SNR=15 dB.

For change detection, we use a 5×5 boxcar to compute the covariance matrices. Fig. 3a shows square-shaped anomalies at 15 dB of SCR compared to the local clutter. As depicted in Fig. 3c, the VAE forces these out-of-distribution data points to be close to the input image pixel distribution, resulting in a considerable energy reduction, thereby facilitating subsequent change detection. Fig. 4 illustrates the quantitative evaluation of all detectors, with $\hat{\Lambda}_{Mat}$ reaches the best overall performance and $\hat{\Lambda}_{MT}$ comes second. Detection maps thresholded at a probability of false alarms (P_{FA}) of 0.5%, calibrated from the H_0 distribution, are shown in Fig. 3d-i and 3h. Note that this value specifies the false-alarm rate

under H_0 and does not correspond to the percentage of activated pixels in the resulting H_1 maps.

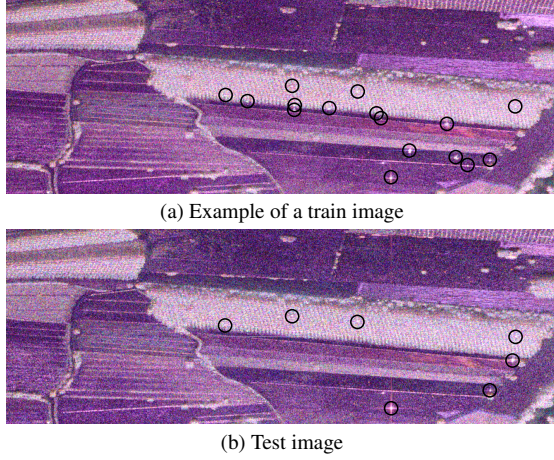


Fig. 5: Dataset image samples. Locations of anomalies are circled in black.

The VAE architecture described in [5] is here used. Nonetheless, since the robust change detectors are derived from the *Complex Compound Gaussian* distribution, we use the $\mathcal{L}_1$ as reconstruction loss. Empirically, setting β to 10^{-7} delivers the lowest validation reconstruction loss and the best ROC curves.

4.2. Experimental Data

The dataset (Fig. 5) consists of fully polarimetric SLC SAR images captured on different dates over a European forest. The dataset comprises six images for training and one image for testing. All images have 444×1258 pixels and were acquired in the L-band at 1.3 GHz, with a ground-range resolution of 1.232 m and a ground-azimuth resolution of 60 cm. Training images are divided into multiple patches of size 64×64 pixels with a stride of 40 pixels, and the size of the latent space z is $q = 16384$. The scene contains anomalies corresponding to corner reflectors and vehicles.

Fig. 6 presents the detection maps obtained in a real test scenario. At the same $P_{fa} = 0.5\%$, $\hat{\Lambda}_{\text{Frobenius}}$ (Fig. 6c), the $\hat{\Lambda}_{\text{RX-SCM}}$ (Fig. 6a) and $\hat{\Lambda}_{\text{RX-Tyler}}$ (Fig. 6b) detect all anomalies but also produces numerous false alarms, especially in the heterogeneous forest area. In Fig. 6d, the $\hat{\Lambda}_{\text{Novak}}$ also recognizes all anomalies but with fewer false alarms in the forest area. The $\hat{\Lambda}_{\text{Mat}}$ doesn't detect all anomalies but yields a substantially cleaner detection map. The $\hat{\Lambda}_{\text{Tex}}$ (Fig. 6f) and $\hat{\Lambda}_{\text{MT}}$ (Fig. 6g) detectors achieve the best trade-off, detecting all seven anomalies while maintaining a reliable detection map comparable to $\hat{\Lambda}_{\text{Mat}}$'s, with $\hat{\Lambda}_{\text{MT}}$ delivering slightly fewer false alarms. These results confirm the advantage of the robust detectors derived under the CCG distribution in modeling SAR images and under the GLRT framework.

5. CONCLUSION

We investigated covariance-equality change detection using robust statistics within a SAR anomaly detection framework based on reconstruction errors. Quantitative results suggest the benefit of robust GLRT-based detectors under the Compound Gaussian model, with the $\hat{\Lambda}_{\text{Mat}}$ detector achieving the best overall performance. In practical scenarios, detector selection remains dependent on the application, operating conditions, and user preferences. Future work will focus on improved learning strategies to further enhance the detection accuracy.

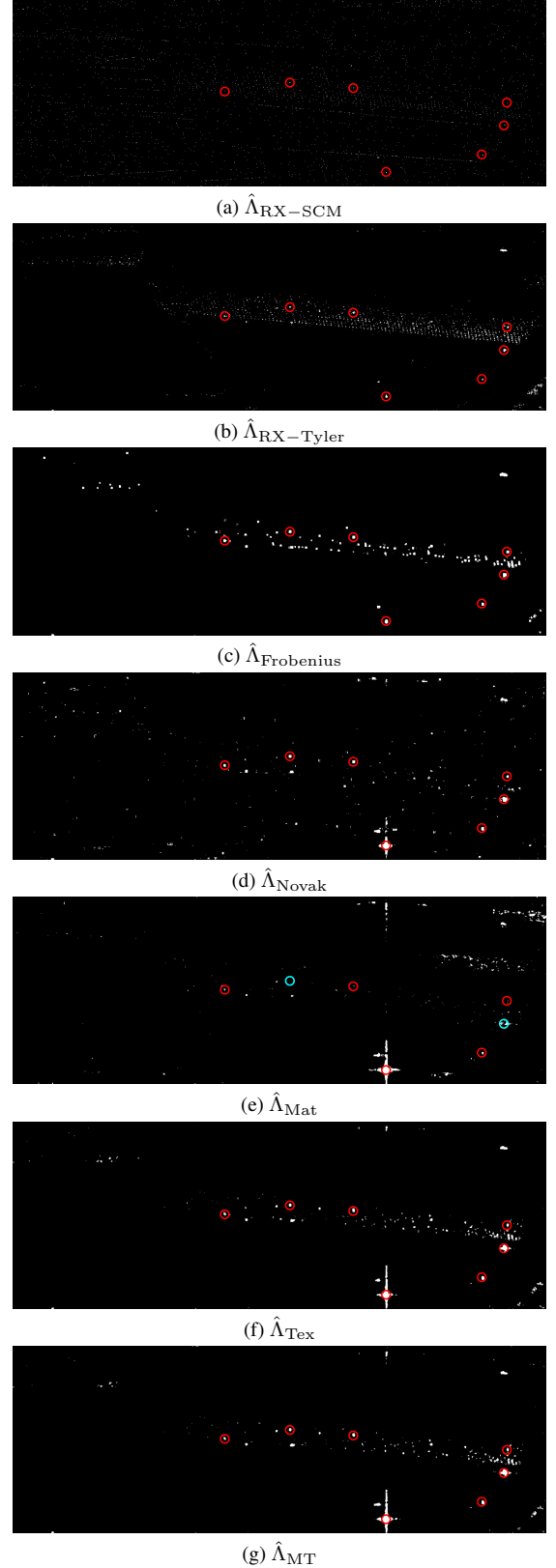


Fig. 6: Detection maps computed on real test image, displayed using the top 0.5% of values. Detected anomalies are circled in red and non-detected in blue.

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